Chapter 5

Quantification of antisite defects in $La_{0.8}Sr_{0.2}MnO_3$ (LSMO) thin films using high angle annular dark field (HAADF) simulations

This chapter presents a framework for quantifying antisite defects in $La_{0.8}Sr_{0.2}MnO_3$ thin films through HAADF-STEM imaging and simulations. It starts by modelling the defects and conducting image simulations, followed by a comparison of experimental images with simulation results to determine the defect density. The study emphasizes the impact of growth conditions on defect density and offers a practical method for defect quantification.

5.1 Introduction and motivation for the study

Complex oxide thin films are attracting significant interest due to their wide-ranging applications in energy and information technologies. These materials are essential in devices like electron and ion conductors [1], photovoltaics [2], thermoelectrics [3], dielectrics [4], and resistive switching systems [5]. They are strongly influenced by grain boundaries, cationic composition, and defect chemistry. Grain boundaries alter electronic and oxygen transport by modifying cation concentrations at interfaces while the local non-stoichiometry in defect chemistry affects electrical resistance, particularly under varying oxygen partial pressures. [6]. Transition metal oxides with ABX_3 perovskite structure can exhibit exceptional properties like superconductivity, magnetoresistance, ionic conductivity, multiferrocity, and piezoelectricity [7–17].

Amongst the perovskite structures, $La_{1-x}Sr_xMnO_3$ (LSMO) has gained significant attention in many research fields due to its unique combination of properties that make it particularly advantageous for various applications. One of the key factors is its low carrier density coupled with a high Curie temperature, which, together with its strong spin polarisation and intrinsic ferromagnetic properties, positions LSMO as an ideal material for spintronic devices [18–24]. The material exhibits a decrease in resistivity upon cooling, linked to a transition from a paramagnetic to a ferromagnetic state [25, 26]. This transition is of particular interest because it leads to a phenomenon known as colossal magnetoresistance, characterised by a substantial negative magnetoresistance at the Curie temperature [27, 19]. Furthermore, the ability to tailor LSMO resistance with low bias makes it highly suitable for applications in tunable radio frequency technologies [28]. It also finds its applications in solid oxide fuel cells [29, 30], upcoming data storage technology, and advanced logic devices[12, 31–36]. Beyond its electronic applications, LSMO nanoparticles have also shown significant potential in the biomedical field such as hyperthermia for cancer treatment [37, 38] and drug delivery systems [39], further establishing its versatility and promise for advancing research across multiple scientific disciplines.

A typical perovskite structure has the formula of ABX_3 where A is usually a large cation, B is a small cation and X is an anion. A sample of $SrTiO_3$ structure is shown in

Fig. 5.1. In the case of LSMO, La and Sr occupy the A site, Mn and O occupy the B and X sites, respectively, as shown in Fig. 5.2. Defects in LSMO are very prominent and are known to have a significant impact on their functionality [40], such as defect-induced spin deterioration [41], and magnetic bi-states phenomena [42] . Amongst the possible defects, antisite defects are a particular type of crystal defect where atoms occupy incorrect lattice positions [7]. In the context of LSMO, antisite defects effectively mitigate the impact of cationic nonstoichiometry on oxygen mass transport properties [43]. Antisite defects affect both, the oxygen diffusivity as well as the concentration of oxygen vacancies [43], making it an important criterion to be studied further. In LSMO, La (A site) occupies a few positions of Mn (B site) as shown in Fig. 5.2. This is because Mn can present multiple valencies (Mn^{3+}, Mn^{4+}), whereas La exhibits only one valence state (La^{3+}) [24]. This allows La to occupy the position of Mn, but not the other way round. Additionally, the high structural stability of Mn facilitates large-scale doping. Owing to this defect, the transport and magnetic properties are highly affected. The defect density can be adjusted to meet specific electrochemical property requirements by tailoring growth conditions, including external pressure, oxygen stoichiometry, and Sr doping [44]. These defects can have low formation energies and are often magnetised, potentially affecting spintronic applications and redox catalysis [7, 45]. Such studies emphasise the crucial role of antisite defects and the need for quantification methods.

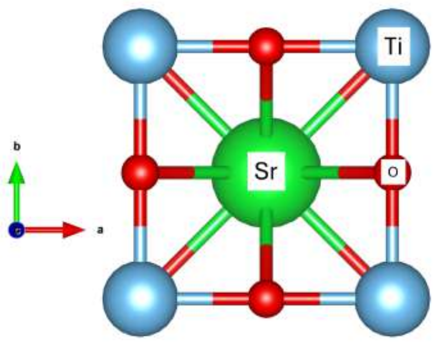


Fig. 5.1 $SrTiO_3$ structure. Sr, Ti, and O are located in the A, B, and X sites of an ABX_3 perovskite respectively.

Antisite defect quantification has been performed for various compounds using a range of methods. For Zr_2Cu, in situ X-ray diffraction coupled with molecular dynamics simulation has been employed to quantify antisite defects during crystallisation [46]. Techniques such as electron spin resonance, infrared absorption, and magnetic circular dichroism have been applied to investigate antisite defects in GaAs [47]. In $LaAlO_3$, temperature-dependent Raman and optical absorption spectroscopies have been utilised for studying these defects [48], while time-resolved synchrotron X-ray diffraction has been used for similar investigations in $LiFePO_4$ [49]. Additionally, Rietveld refinement analysis has proven effective in examining antisite defects in $LiNiPO_4$ [50]. High-angle annular dark-field (HAADF) and annular bright-field (ABF) in scanning transmission electron microscopy (STEM) have enabled the direct visualisation of antisite defects in $LiCoPO_4$ nanocrystals [51]. Furthermore, efforts have been made to observe, control, and quantify antisite defects in several oxides, including $Li_{1.2}Ni_{0.2}Mn_{0.6}O_2$ [52], Sr_2FeMoO_6 [53] and $Y_3Al_5O_{12}$ [54] [55]. However, there have been limited attempts to quantify antisite defects in LSMO.

HAADF-STEM in aberration-corrected microscopes enables atomic-resolution Z-contrast imaging, where the image intensity scales with the atomic number (Z) raised to an exponent α, typically ranging from 1.4 to 2 [56, 57]. Heavy elements appear noticeably brighter than lighter ones due to incoherent electron scattering at high angles, which is the primary cause of contrast in HAADF images. This makes it easier to distinguish and identify atomic columns in complex materials [58]. Atomic number, specimen thickness, and microscope parameters like detector angles are some of the variables that affect HAADF contrast [59, 60]. When detector angles (and other parameters) are fixed, increasing the sample thickness leads to higher background intensity, which needs to be subtracted for better accuracy in quantitative analysis [59]. In the case of LSMO, the differences in atomic number between La (Z=57), Sr (Z=38), and Mn (Z=25) make HAADF-STEM an ideal tool for detecting these defects in LSMO. Changes in the composition should affect the intensity of the atomic columns in the HAADF images. Using this principle, site occupancies and defect concentrations have been studied, particularly in systems where the local intensity is modulated by antisite substitutions or mixed occupancies. For instance, Chung et al. [61] directly visualised antisite defects in $LiFePO_4$ using atomic-resolution HAADF imaging and multislice simulations, comparing measured intensity variations with modelled occupancy levels. Similarly, in order to measure cation column occupancy and identify light-element vacancies, Findlay et al. [62] quantified the HAADF intensity in $SrTiO_3$ and $BaTiO_3$. They were able to determine occupancy factors with a few percent accuracy by comparing experimental profiles with frozen phonon simulations. Further, Van Aert et al. [63] used HAADF contrast imaging combined with statistical parameter estimation to quantify atom counts in GaN and AlN columns by mapping calibrated atomic column intensities. While HAADF-STEM has been successfully applied to study antisite defects in other materials [7, 61, 64], its application to LSMO, particularly with respect to the quantification of the antisite defect, has been unexplored.

Here we propose a method for the quantification of antisite defects in experimental HAADF images through image simulation. Further, we use the methodology to quantify the defects in the experimental LSMO samples fabricated under different preparation conditions, leading to variable amounts of antisite defects. Our study aims to establish a method for correlating experimental observations with defect quantities, thereby facilitating more accurate and comprehensive analysis of antisite defects in large regions of a given specimen.

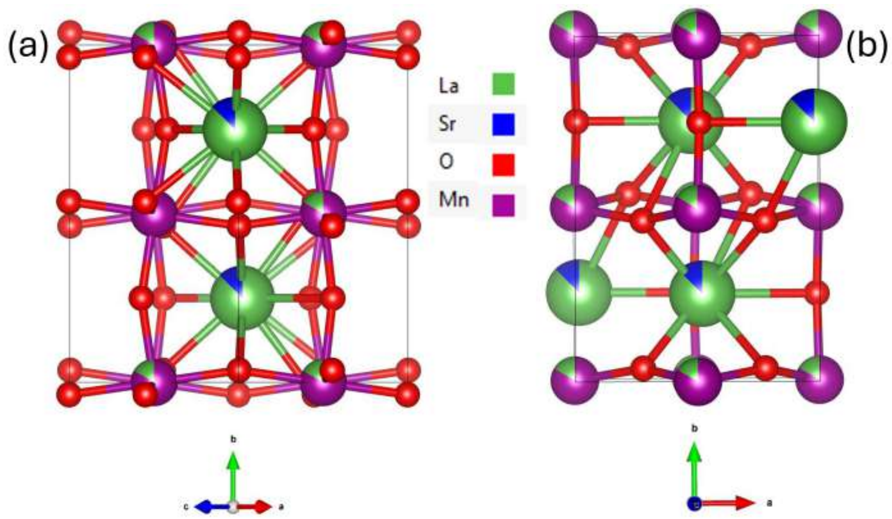


Fig. 5.2 Orthorhombic unit cell of LSMO in two orientations (a) [101] & (b) [001].

5.2 Methodology

5.2.1 Experimental sample preparation

$La_{0.8}Sr_{0.2}Mn_yO_{3\pm\delta}$ samples were grown at the Catalonia Institute for Energy Research (IREC), by the group of Albert Tarancon and co workers. The films were grown on $SrTiO_3$ (001) single crystal substrates (Crystec GmbH) by Pulsed laser Deposition (PLD) using a large area PLD system (PVD Systems-PLD 5000) equipped with a 248 nm KrF excimer laser (Lambda Physics-COMPex PRO 205). LSM_y thin films were grown at 700 ºC with an energy fluence of 1 J cm^{-2} per pulse, a frequency of 10 Hz, and a substrate-target distance of 95 mm. The Mn/(La + Sr) ratio was varied by controlling the background pressure during deposition. Indeed, as demonstrated in our previous work [43], an oxygen background pressure of $pO_2 = 2.6 \times 10^{-2}$ mbar can be used to grow nearly stoichiometric LSM layers ($y \sim 1.00$), while highly Mn-deficient thin films ($y \sim 0.85$) are obtained for $pO_2 = 6.6 \times 10^{-3}$ mbar. The samples were in-situ annealed in 0.3 mbar of oxygen at 700 ºC after deposition to compensate for oxygen deficiency in the layers.

5.2.2 Image simulation

Apart from the antisite defects ($La^{\times}_{Mn}$), LSMO also contains various types of vacancies and defects, they include B-site vacancies (V'''_{Mn}), A-site vacancies (V'''_{La}), oxygen vacancies ($V^{\bullet\bullet}_{O}$), and positively charged defects ($Mn^{\bullet}_{Mn}$), as reported in [43]. The concentration of these defects and vacancies is illustrated in the Brouwer diagram shown in Fig. 5.3. At a working pressure of 0.3 mbar, the concentration of $La^{\times}_{Mn}$ is approximately three orders of magnitude higher than that of V'''_{Mn} and V'''_{La}, and about seven orders of magnitude higher than that of $V^{\bullet\bullet}_{O}$. Since the focus of this study is on analysing the intensity of atomic columns in HAADF images, only the $La^{\times}_{Mn}$ defect has been considered, as the contribution of the other defects is expected to be minimal in the imaging results.

To simulate the antisite defects, LSMO crystallographic structures were atomically modelled for HAADF image simulation with varying amounts of antisite defects only, (as shown in Table 5.1). The value of y is a measure of the ratio of the total amount of Mn in the sample to the sum of the total amount of (La and Sr) in the A site and the amount of La in the B site of the sample, as shown in equation 5.1. The crystallographic modelled structures for two different zone axes, [101] and [001] are shown in Fig. 5.2. The fractional occupancy of the La atom in Mn atomic columns (i.e. A site atoms in B columns) depicts the antisite defects and represents the average composition along the atomic columns in the case of image simulation. Similar models were designed for each value of y .

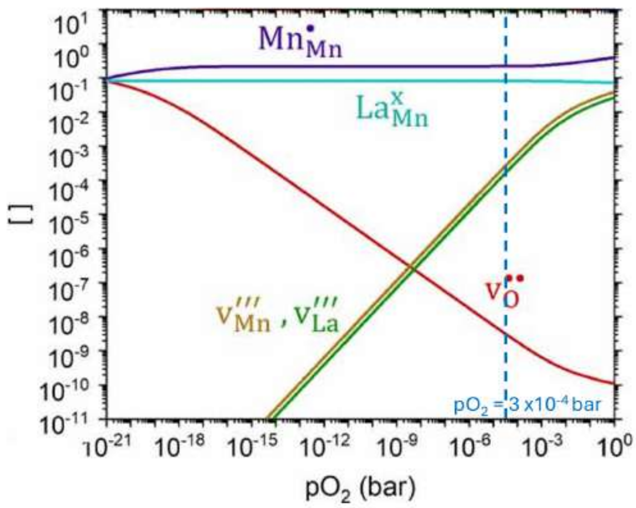


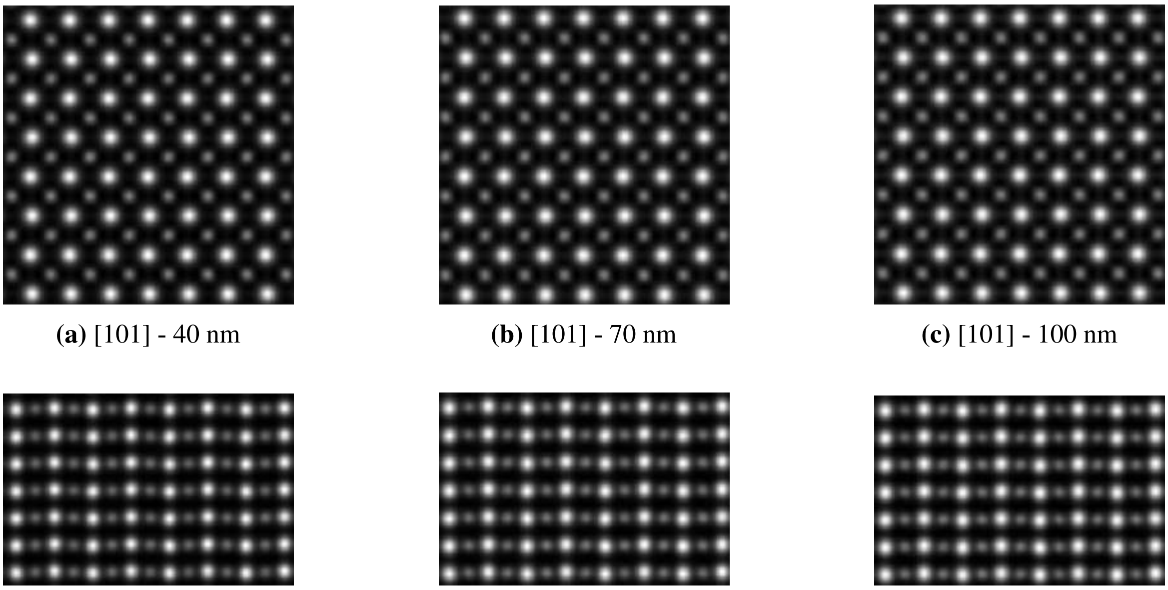
Fig. 5.3 Brouwer diagram representing the concentration of various defects and vacancies in LSMO with y = 0.85, adapted from [43].

The STEM HAADF image simulations were conducted using the multislice algorithm of Dr. Probe [65] for the zone axes of [101] & [001]. The thickness of the sample was set in the range of 50-100 nm with an increment of 10 nm in each simulation. The acceleration voltage was set at 200 kV and the convergence semiangle at 24.1 mrad. The HAADF detector was setup with an outer angle of 250 mrad and an inner angle of 80 mrad. Along the z direction, the number of slices and variants per slice per unit cell was set at 4 and 100 respectively. An example of HAADF image simulation with y = 0.88 is shown in Fig. 5.4.

$$y = \frac{\text{Total amount of Mn in the sample}}{\text{Total amount of [(La + Sr) in A site + La in B site]}} \tag{5.1}$$

y	La (%)	Mn (%)
0.78	12.4	87.6
0.80	11.1	88.9
0.85	8.1	91.9
0.88	6.4	93.6
0.92	4.2	95.8
0.94	3.1	96.9
0.99	0.5	99.5

Table 5.1 Composition of B site in LSMO with varying y.



(a) [101] - 40 nm **(b)** [101] - 70 nm **(c)** [101] - 100 nm

(d) [001] - 40 nm **(e)** [001] - 70 nm **(f)** [001] - 100 nm

Fig. 5.4 HAADF image simulation of LSMO with $y = 0.88$ for two zone axes and different sample thickness. The brighter white columns correspond to A site and the less bright to B (defective column) site.

Once the HAADF images were simulated, the column intensities of A and B site were integrated using the ATOMAP [66] software, by using the Voronoi method [67] to integrate signal around the atomic columns in the lattice. The radius of each column was measured to account for the intensity from the corresponding atomic column while avoiding contributions from neighboring columns. In order to account for the background signal, a background intensity was also calculated in between two A sites as shown in Fig. 5.5. The radius of this was chosen as half of that of the A and B sites, while the radius of A and B were kept identical.

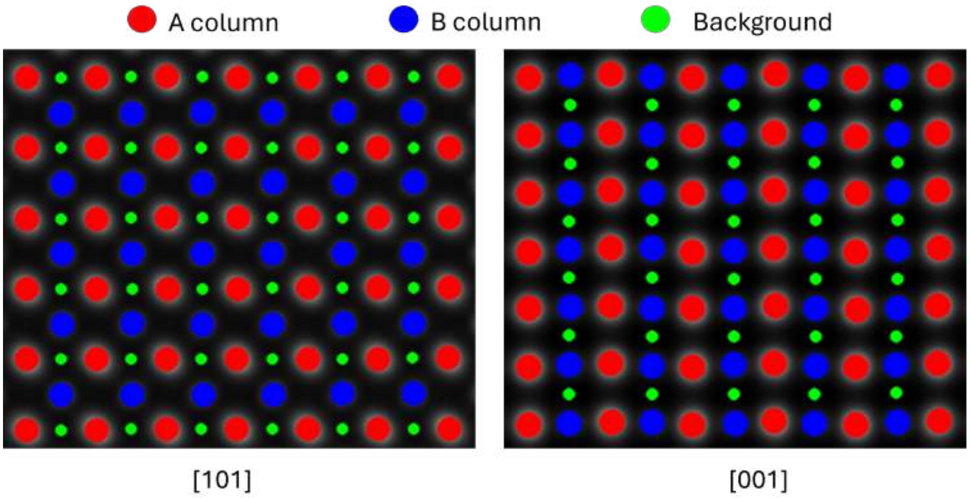


Fig. 5.5 Schematic representation of A, B, and background sites used for HAADF image analysis. The site intensities were integrated using the Voronoi method in ATOMAP.

For each simulation, these intensities of columns A, B and the background were measured and a relative intensity ratio (RIR) parameter was calculated as follows.

$$\text{RIR (Relative Intensity Ratio)} = \frac{(\text{Avg. A site intensity} - \text{Avg. B site intensity}) \times 100}{\text{Avg. A site intensity} - \text{ Background intensity} \times 4}$$

Background signal is multiplied by the square of the integration radius ratio (here 2^2) in order to have the same area as the integrated A and B positions. This ratio takes into account the change in intensity of B site due to the presence of the antisite defects by taking the A site intensity as a reference.

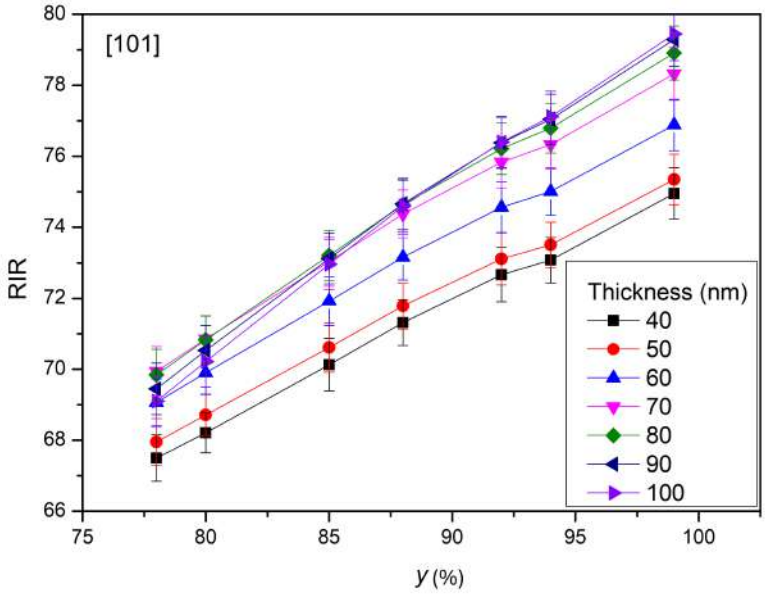
Following which, we implemented the ATOMAP A, B and background intensity integration on the experimental images, calculated their RIR and compared that with our simulation to evaluate the density of antisite defect in our experimental sample.

5.3 Simulation results

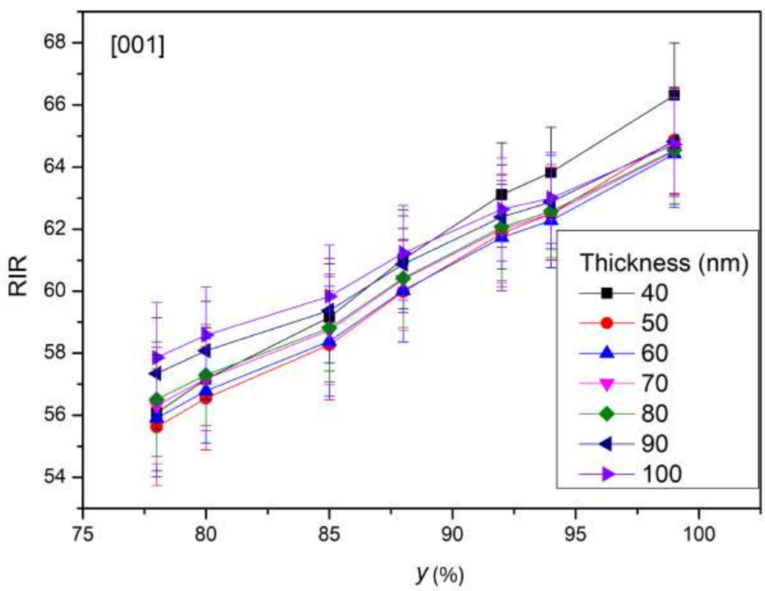
The intensity of the columns in a HAADF image depends on the thickness, atomic number and the density of the atoms in that particular column. The dependency of RIR on thickness was evaluated by doing the HAADF simulations from 40 nm to 100 nm in steps of 10 nm for two zone axes [101] and [001]. The results are shown in Fig. 5.6. This plot reveals that the RIR is dependent on the thickness of the material. The change in the differences in the RIR of the two zone axes for the same thickness can be attributed to the fact that in the zone axis [001] the effective number of La and Mn atoms are lower than that of [101] and with higher thickness, the difference in the number of atoms increases as well, affecting the RIR. However, in general, it can be concluded that as the value of y increases, the RIR also increases. This chart serves as the basis for the quantification of antisite defects from the experimental images.

5.4 Experimental results

Two samples were studied, one fabricated with an oxygen partial pressure of 2 mTorr and another with 5 mTorr. These will hereafter be referred to as the 2 mT and 5 mT samples, respectively. The cross sections of the two samples were prepared in the standard FIB procedure and along the [101] zone axis. The lamellae were then observed in a FEI Titan 3 Themis 60-300 microscope operated at 200 kV. The HAADF images, along with its corresponding experimental FFT and the theoretical diffraction pattern are shown in Fig. 5.7. In both cases, the A and B sites are well resolved. EELS log-ratio relative thickness determination was also carried out on low-loss spectra to calculate the thickness of the samples. The t/λ ratio spread across the sample is shown in Fig. 5.8. It can be observed that the t/λ is quite uniformly spread throughout the sample with an average value of 0.57 for 2 mT and 0.70 for 5 mT with a standard deviation of 0.01 and 0.03 respectively. It also reveals the fact that our samples have uniform thicknesses. The mean free path was calculated by following the methodology proposed by Lakoubovskii et al. [68] and Egerton [69] and was found to be 109.142 nm. Using the t/λ ratio and the mean free path, the thickness of the sample was calculated. The thickness of the 2mT and 5mT samples was calculated to be 62 nm and 76 nm respectively, with standard deviations of 1 nm and 3 nm.

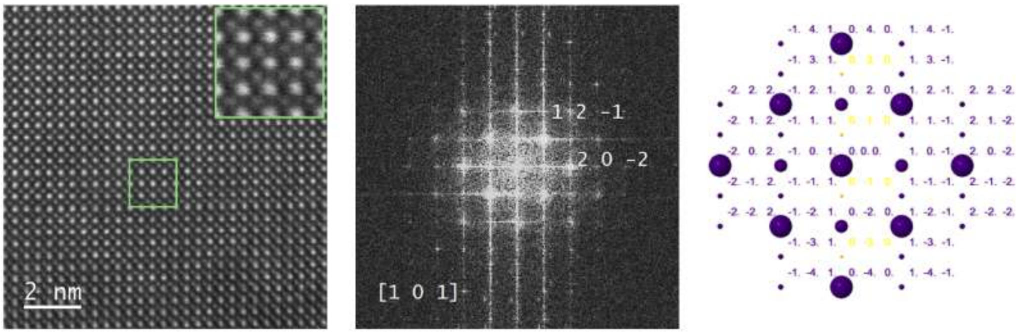


(a)

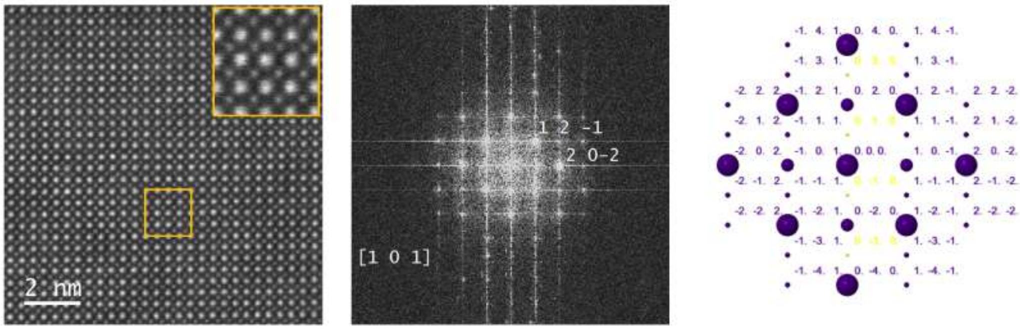


(b)

Fig. 5.6 Variation of RIR as a function of thickness of the sample and *y*, for two zone axes (a) 101 and (b) 001.

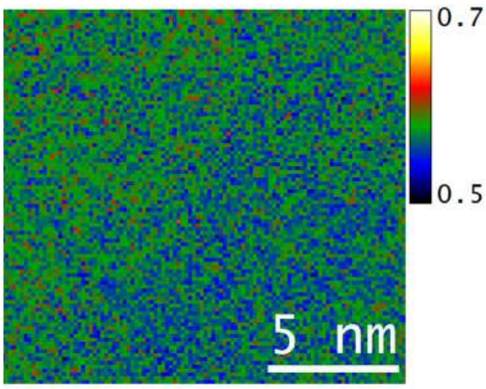


(a) 2mT

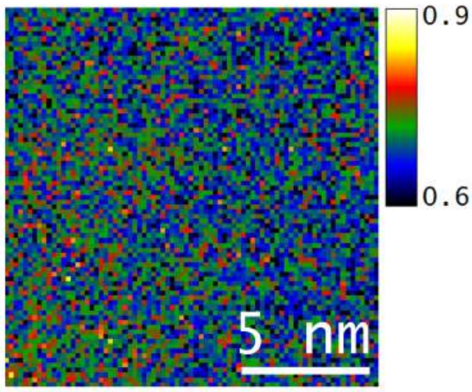


(b) 5mT

Fig. 5.7 HAADF experimental image (left) with its corresponding FFT (centre) and the theoretical diffraction pattern (right) of (a) 2mT and (b) 5mT. The analysis reveals that the experimental samples have been observed along [101] zone axis.



(a) 2mT [101]



(b) 5mT [101]

Fig. 5.8 t/λ distribution across the sample for (a) 2mT [101] and (b) 5mT [101]. The distribution is well spread, confirming the uniform thickness of the samples.

5.5 Antisite quantification on experimental results

ATOMAP analysis was performed on all the HAADF experimental images for each sample. The A and B columns were identified as well as the background intensity site, as shown in Fig. 5.9 for the sample of 2mT, and their integrated intensities were evaluated. Subsequently, the RIR was calculated for each HAADF image of a particular sample, and the average RIR was determined. The average RIR value of 2mT and 5mT was evaluated to be 73.7 ± 1.4 and 75.9 ± 0.8 respectively.

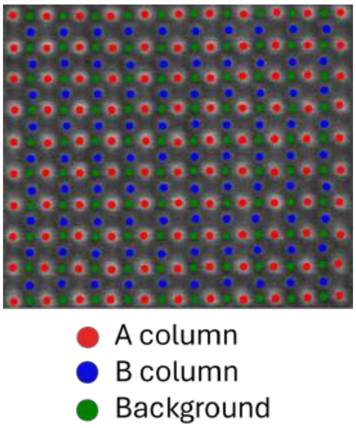


Fig. 5.9 Representation of A, B and Background intensity sites on the experimental HAADF image of 2mT in [101] direction.

Once the thicknesses of the experimental samples were determined, the relationship between RIR and y for each specific thickness needed to be established for defect quantification. To achieve this, the simulated data presented in Fig. 5.6(a) was used to interpolate the corresponding plots for 62 nm and 76 nm thicknesses. Linear interpolation was performed using the data for 50-60-70 nm thicknesses to obtain the 62 nm profile, and the 70-80-90 nm thicknesses to derive the 76 nm profile. The resulting RIR vs. y profiles are shown in Fig. 5.10. It can be observed that the plots in this graph can be fit well with a linear equation.

To evaluate the antisite defect quantification in the samples, the previously obtained RIR values for 2mT (62 nm thick) and 5mT (76 nm thick) from the experimental HAADF images were fitted into the linear equation derived in Fig. 5.10, yielding the corresponding y values, which represent the defect composition. The evaluated y (Δy) values for 2mT and 5mT were found to be 0.90 (0.04) and 0.92(0.04), respectively.

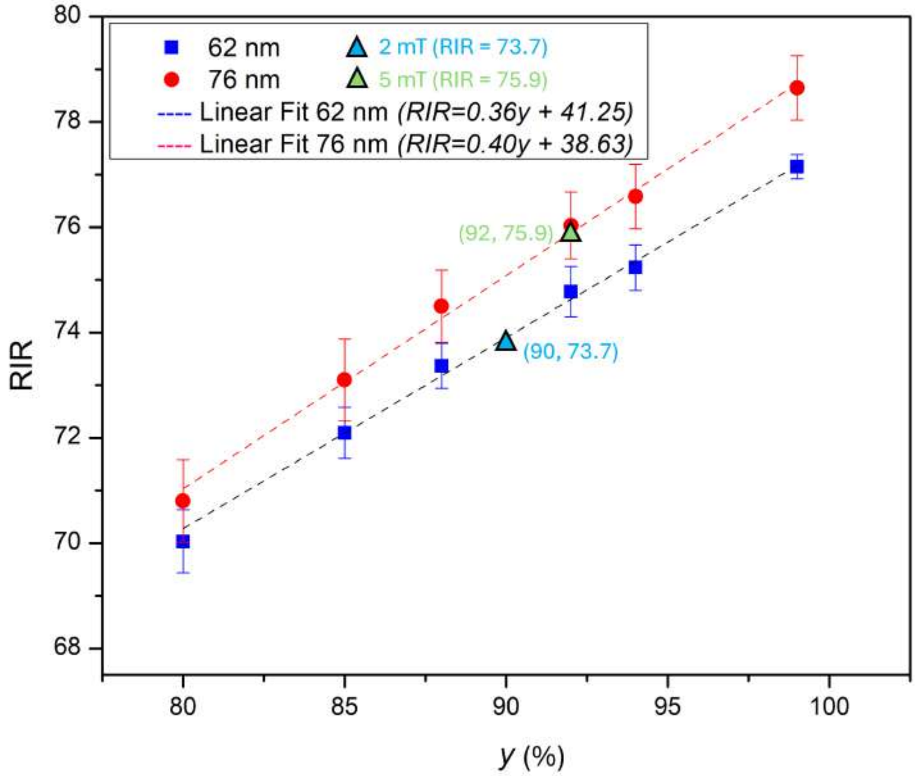


Fig. 5.10 Plot of RIR vs y for the thicknesses of 62 nm and 76 nm by interpolating the data from the Fig. 5.6(a) along with the fit of experimental RIR values of 2mT and 5mT on its corresponding thickness plot of 62 nm and 76 nm respectively.

By correlating y with the B-site composition, it can be determined that the 2mT sample consists of 94.5% Mn and 5.5% La at the B-site. Similarly, for the 5mT sample, the B-site composition is 96% Mn and 4% La. These results are summarised in Table 5.2. Based on these findings, it can be concluded that the sample fabricated under 5mT pressure exhibits a lower concentration of antisite defects compared to the 2mT sample. The observed changes in Mn composition are approximately 1% of the total composition. Under ideal circumstances and with the right adjustments, quantification error in EELS normally falls between 1% and 2%, but it can rise to 10% to 30% or more based on variables like sample thickness, background subtraction precision, and instrumental parameters like collection angle and alignment [70–73, 69, 74]. Our study demonstrates that the used procedure performs a quantification of antisite defects beyond the standard TEM analytical techniques. This approach facilitates the quantification of antisite defects whenever atomically resolved HAADF images are available and can be applied to similar samples, allowing defect quantification in future studies.

Sample	**RIR**	**Thickness**	***y* (Δ*y*)**	**B site composition (%)**	
				La	**Mn**
2 mT	73.7	62 nm	0.90 (0.04)	5.5	94.5
5 mT	75.9	76 nm	0.92 (0.04)	4	96

Table 5.2 Quantification of antisite defects of the experimental samples, displaying their RIR, thickness, *y* values, and B site composition.

5.6 Discussion

This chapter introduces an approach in quantitative defect analysis for advanced oxide materials with a focus on LSMO systems.

The innovation lies in the use of relative HAADF-STEM intensities as a strategy for quantifying antisite defects. By determining the interrelation between RIR, sample thickness, and defect concentration using systematic simulations, we offer a framework that can be applied to other perovskite systems with analogous defect structures. The correlation between growth oxygen partial pressure and antisite defect concentration as observed serves as direct proof that the chemistry of defects is significantly affected by synthesis conditions.

Although the emphasis on $La_{Mn}^{\times}$ antisite defects is well founded because they have a majority concentration, the existence of other defect species (vacancies, interstitials) adds some indeterminacy to the interpretation. The hypothesis that other defects have no significant influence on intensity fluctuations might not be universally valid for all synthesis conditions or perovskite compositions.

The need for accurate thickness determination via EELS t/λ measurements increases the analysis workflow complexity but is necessary for reliable quantitation. The relatively homogeneous thickness distributions found in our samples may not be obtainable in all specimen preparation protocols, which could restrict the applicability of the method.

For solid oxide fuel cell applications of LSMO, quantifying antisite defects is critical for ionic conductivity and catalytic optimisation. The phenomenon of synthesis pressure versus defect density offers a clear path for material property tuning via processing parameter control. Application to two distinct samples proves its usefulness, yet more general validation across laboratories and different specimen preparation routes will enhance confidence in the general use of the methodology.

5.7 Conclusions

Overall, the findings of the study can be summarised as the following.

- HAADF-STEM imaging was employed to detect and quantify antisite defects. This technique provides Z-contrast imaging, enabling to measure the change in intensity of the image with respect to effective atomic number in a column.
- HAADF image simulations of LSMO samples for [101] and [001] zone axes were performed with varying defect quantity and sample thicknesses.
- Relative Intensity Ratio (RIR) parameter was evaluated which provided a framework for the analysis of experimental images.
- Experimental HAADF images were analysed using ATOMAP software to measure the RIR. Comparison with simulation data allowed for the quantification of the defect in the experimental sample.
- The developed methodology provides a first-hand approximation for quantifying antisite defects in LSMO, with potential applicability to other perovskite materials.

5.8 References

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